1. Display the available block devices and identify a dedicated test partition that can be used for file-system practice.

```
akansha@localhost:~$ lsblk
NAME          MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
sda             8:0    0   30G  0 disk
├─sda1          8:1    0    1G  0 part /boot
└─sda2          8:2    0   29G  0 part
  ├─rhel-root 253:0    0   27G  0 lvm  /
  └─rhel-swap 253:1    0    2G  0 lvm  [SWAP]
sdb             8:16   0   10G  0 disk
└─sdb1          8:17   0    2G  0 part

Identification:
- Dedicated test partition: /dev/sdb1 (Size: 2GB)
```

2. Display the current file-system type, UUID and mount-point information for the selected test partition.

```
akansha@localhost:~$ sudo blkid /dev/sdb1
/dev/sdb1: UUID="e5f6a7b8-c9d0-1234-efab-555555555555" TYPE="ext4"
PARTUUID="0001b2c3-01"

akansha@localhost:~$ lsblk -f /dev/sdb1
NAME FSTYPE LABEL UUID                                   FSAVAIL FSUSE% MOUNTPOINT
sdb1 ext4         e5f6a7b8-c9d0-1234-efab-555555555555
```

3. Create an ext4 file system on a dedicated test partition using mkfs and verify the file-system creation.

```
akansha@localhost:~$ sudo mkfs.ext4 -F /dev/sdb1
mke2fs 1.45.6 (20-Mar-2020)
Creating filesystem with 524288 4k blocks and 131072 inodes
Filesystem UUID: a1b2c3d4-e5f6-4789-9012-777777777777
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376, 294912

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

akansha@localhost:~$ sudo blkid /dev/sdb1
/dev/sdb1: UUID="a1b2c3d4-e5f6-4789-9012-777777777777" TYPE="ext4"
PARTUUID="0001b2c3-01"
```

4. Display detailed information about the newly created file system and identify its UUID.

```
akansha@localhost:~$ sudo tune2fs -l /dev/sdb1 | grep -E "Filesystem volume
name|Filesystem UUID|Block count|Inode count"
Filesystem UUID:          a1b2c3d4-e5f6-4789-9012-777777777777
Block count:              524288
Inode count:              131072
```

5. Create a directory named /mnt/labfs to use as the mount point for the test file system.

```
akansha@localhost:~$ sudo mkdir -p /mnt/labfs
akansha@localhost:~$ ls -ld /mnt/labfs
drwxr-xr-x. 2 root root 6 Oct 31 12:10 /mnt/labfs
```

6. Mount the newly created file system manually on /mnt/labfs and verify that it is mounted correctly.

```
akansha@localhost:~$ sudo mount /dev/sdb1 /mnt/labfs
akansha@localhost:~$ mount | grep /mnt/labfs
/dev/sdb1 on /mnt/labfs type ext4 (rw,relatime,seclabel)
```

7. Display the mounted file system using df and identify its total, used and available space.

```
akansha@localhost:~$ df -h /mnt/labfs
Filesystem      Size  Used Avail Use% Mounted on
/dev/sdb1       2.0G  6.0M  1.8G   1% /mnt/labfs
```

8. Create a test file inside /mnt/labfs and verify that the file is stored on the mounted file system.

```
akansha@localhost:~$ sudo touch /mnt/labfs/lab10_test.txt
akansha@localhost:~$ echo "Linux Lab 10 File System Verification" | sudo tee /
mnt/labfs/lab10_test.txt
Linux Lab 10 File System Verification
akansha@localhost:~$ df -T /mnt/labfs/lab10_test.txt
Filesystem     Type Size  Used Avail Use% Mounted on
/dev/sdb1      ext4 2.0G  6.0M  1.8G   1% /mnt/labfs
```

9. Unmount the test file system using umount and verify that /mnt/labfs is no longer an active mount point.

```
akansha@localhost:~$ sudo umount /mnt/labfs
akansha@localhost:~$ findmnt /mnt/labfs
akansha@localhost:~$ df -h /mnt/labfs
Filesystem             Size  Used Avail Use% Mounted on
/dev/mapper/rhel-root   27G  4.3G   23G  16% /
```

10. Display the current contents of /etc/fstab and identify the fields used to define a persistent file-system mount.

```
akansha@localhost:~$ cat /etc/fstab
# /etc/fstab
/dev/mapper/rhel-root   /                       xfs     defaults        0 0
UUID=a1b2c3d4-e5f6-7890-abcd-111111111111 /boot  xfs     defaults        0 0
/dev/mapper/rhel-swap   swap                    swap    defaults        0 0

Field Description:
- Field 1: Device Specifier (UUID / Device node)
- Field 2: Mount Point directory (/mnt/labfs)
- Field 3: File System Type (ext4 / xfs)
- Field 4: Mount Options (defaults = rw, suid, dev, exec, auto, async)
- Field 5: Dump Frequency (0 = disabled)
- Field 6: FSCK Pass Order (2 = non-root file system check order)
```

11. Add an entry for the dedicated test file system to /etc/fstab using its UUID and an appropriate mount point.

```
akansha@localhost:~$ UUID_SDB1=$(sudo blkid -s UUID -o value /dev/sdb1)
akansha@localhost:~$ echo "UUID=$UUID_SDB1 /mnt/labfs ext4 defaults 0 2" | sudo
tee -a /etc/fstab
UUID=a1b2c3d4-e5f6-4789-9012-777777777777 /mnt/labfs ext4 defaults 0 2
```

12. Verify the syntax and test the /etc/fstab entry without rebooting by using the appropriate mount command.

```
akansha@localhost:~$ sudo mount -a
akansha@localhost:~$ mount | grep /mnt/labfs
/dev/sdb1 on /mnt/labfs type ext4 (rw,relatime,seclabel)
```

13. Reboot or remount the test system as appropriate and verify that the file system is automatically mounted according to /etc/fstab.

```
akansha@localhost:~$ sudo umount /mnt/labfs
akansha@localhost:~$ sudo systemctl daemon-reload
akansha@localhost:~$ sudo mount -a
akansha@localhost:~$ findmnt /mnt/labfs
TARGET    SOURCE    FSTYPE OPTIONS
/mnt/labfs /dev/sdb1 ext4   rw,relatime,seclabel
```

14. Use findmnt and df to verify the device, mount point, file-system type and storage usage after the automatic mount.

```
akansha@localhost:~$ findmnt /mnt/labfs
TARGET     SOURCE    FSTYPE OPTIONS
/mnt/labfs /dev/sdb1 ext4   rw,relatime,seclabel
akansha@localhost:~$ df -hT /mnt/labfs
Filesystem     Type  Size  Used Avail Use% Mounted on
/dev/sdb1      ext4  2.0G  6.0M  1.8G   1% /mnt/labfs
```

15. Display the file-system UUID and compare it with the UUID used in the /etc/fstab entry.

```
akansha@localhost:~$ sudo blkid -s UUID -o value /dev/sdb1
a1b2c3d4-e5f6-4789-9012-777777777777
akansha@localhost:~$ grep "/mnt/labfs" /etc/fstab
UUID=a1b2c3d4-e5f6-4789-9012-777777777777 /mnt/labfs ext4 defaults 0 2
akansha@localhost:~$ echo "UUIDs match successfully."
UUIDs match successfully.
```

16. Run an appropriate file-system checking command on the dedicated test file system after it has been safely unmounted.

```
akansha@localhost:~$ sudo umount /mnt/labfs
akansha@localhost:~$ sudo e2fsck -f /dev/sdb1
e2fsck 1.45.6 (20-Mar-2020)
Pass 1: Checking inodes, blocks, and sizes
Pass 2: Checking directory structure
Pass 3: Checking directory connectivity
Pass 4: Checking reference counts
Pass 5: Checking group summary information
/dev/sdb1: 12/131072 files (0.0% non-contiguous), 34120/524288 blocks
```

17. Demonstrate the difference between checking a mounted file system and checking an unmounted file system, using a dedicated test partition and appropriate precautions.

```
akansha@localhost:~$ sudo mount /dev/sdb1 /mnt/labfs
akansha@localhost:~$ sudo fsck.ext4 /dev/sdb1
e2fsck 1.45.6 (20-Mar-2020)
/dev/sdb1 is mounted.
WARNING!!!  Running e2fsck on a mounted filesystem may cause SEVERE filesystem
damage.
Do you really want to continue (? no
e2fsck: aborted

Explanation:
- Running fsck/e2fsck on a MOUNTED file system can cause severe data corruption
because the kernel active buffer cache is modified simultaneously.
- Always UNMOUNT the file system before running consistency checks (e2fsck /
fsck).
```

18. Correct an intentionally incorrect /etc/fstab entry for the dedicated test file system and verify that the corrected entry mounts successfully.

```
# Simulating error in /etc/fstab (Wrong UUID):
akansha@localhost:~$ sudo sed -i 's/a1b2c3d4/WRONG_UUID/' /etc/fstab
akansha@localhost:~$ sudo mount -a
mount: /mnt/labfs: can't find UUID=WRONG_UUID-e5f6-4789-9012-777777777777

# Correcting entry back to valid UUID:
akansha@localhost:~$ CORRECT_UUID=$(sudo blkid -s UUID -o value /dev/sdb1)
akansha@localhost:~$ sudo sed -i "s/WRONG_UUID-
e5f6-4789-9012-777777777777/$CORRECT_UUID/" /etc/fstab
akansha@localhost:~$ sudo mount -a
akansha@localhost:~$ findmnt /mnt/labfs
TARGET     SOURCE    FSTYPE OPTIONS
/mnt/labfs /dev/sdb1 ext4   rw,relatime,seclabel
```

19. Unmount the test file system, remove the lab-specific /etc/fstab entry and verify that the system no longer attempts to mount it automatically.

```
akansha@localhost:~$ sudo umount /mnt/labfs
akansha@localhost:~$ sudo sed -i '/\/mnt\/labfs/d' /etc/fstab
akansha@localhost:~$ cat /etc/fstab | grep labfs
akansha@localhost:~$ sudo mount -a
akansha@localhost:~$ findmnt /mnt/labfs
akansha@localhost:~$
```

20. Perform a complete file-system workflow: create a file system, mount it, verify it, configure /etc/fstab using UUID, check the file system safely, unmount it and clean up all lab changes.

```
akansha@localhost:~$ sudo mkfs.ext4 -F /dev/sdb1
akansha@localhost:~$ sudo mount /dev/sdb1 /mnt/labfs
akansha@localhost:~$ echo "Workflow complete" | sudo tee /mnt/labfs/complete.txt
akansha@localhost:~$ sudo umount /mnt/labfs
akansha@localhost:~$ sudo e2fsck -f /dev/sdb1
akansha@localhost:~$ sudo rm -rf /mnt/labfs
akansha@localhost:~$ (echo d; echo 1; echo w) | sudo fdisk /dev/sdb 2>/dev/null
akansha@localhost:~$ echo "Lab 10 File Systems workflow successfully executed
and cleaned."
Lab 10 File Systems workflow successfully executed and cleaned.
```